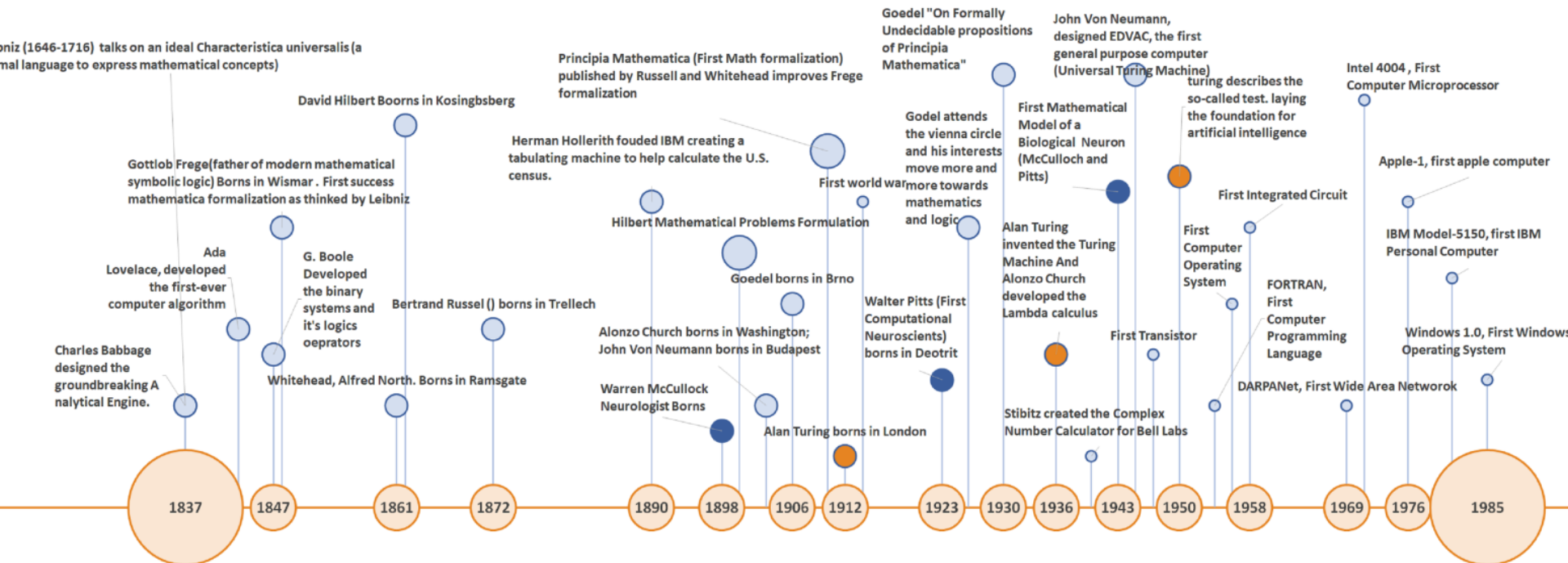


Storia della Computazione

Da Leibniz a ChatGPT





Alan Turing

Padre dell'Informatica moderna

- Tra i tanti citiamo *semplici* suoi contributi:

1936: *On computable numbers with an application to the entscheidungsproblem*, The Graduate College, Princeton University, New Jersey.

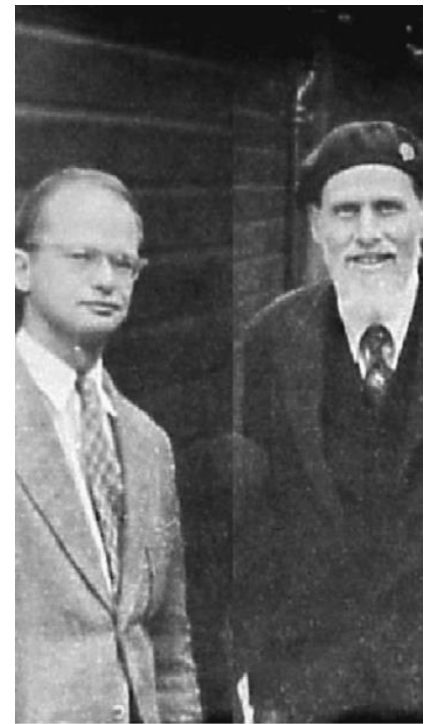
1950: *Computing Machinery and Intelligence*, Mind, 1950.

- Nel primo lavoro a solo 23 anni, definisce le basi teoriche del concetto di '*calcolo automatico*'.
- Definisce una Macchina per il Calcolo come astrazione formale. (a programma cablato)
- Dimostra l'esistenza di una Macchina UNIVERSALE.
- Dimostra l'esistenza di problemi NON Calcolabili.
- Introduce il Test di Turing, come gioco dell'imitazione per comprende una macchina è *Intelligente*.

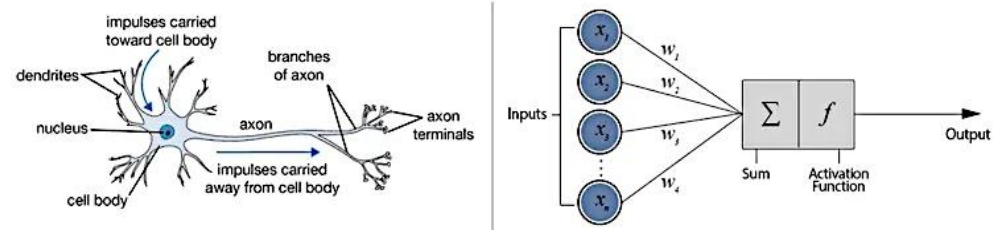


1943 - Warren McCulloch, Walter Pitts

- Neurophysiologist and cybernetician of the University of Illinois at Chicago: Warren McCulloch and self-taught logician and cognitive psychologist Walter Pitts published:
 - “*A Logical Calculus of the ideas Imminent in Nervous Activity*”
 - describing the “McCulloch - Pitts *artificial neuron*” the first mathematical model of a neural network.
- Building on ideas in Alan Turing’s they provided a way to describe brain functions in abstract terms.
- They showed that simple elements connected in a neural network can have immense computational power.
- The paper received little attention until its ideas were applied by John von Neumann, Norbert Wiener, and others.



Biological Neuron versus Artificial Neural Network



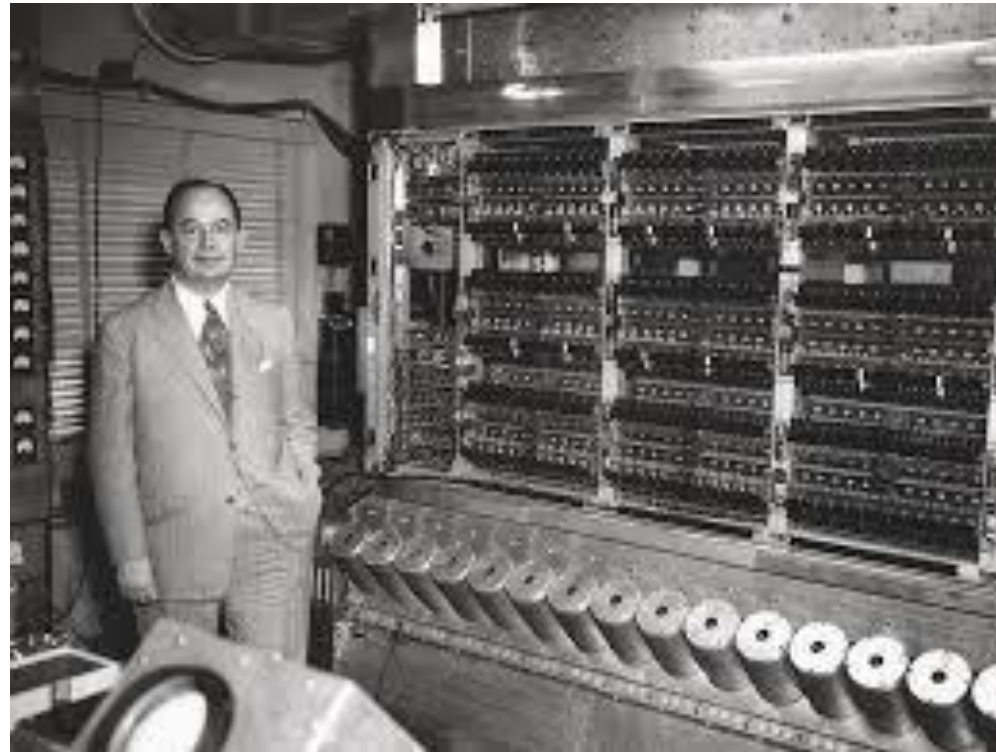
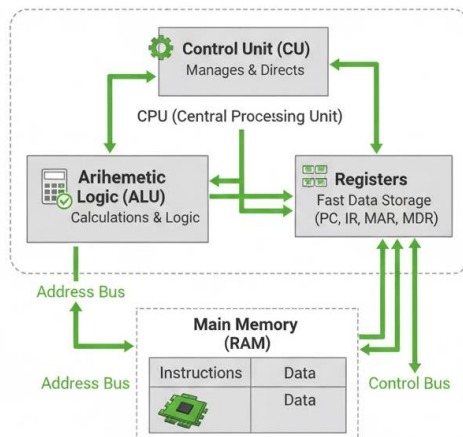
1945

John von Neumann writes *First Draft of a Report on the EDVAC*

- Definisce le componenti dell'architettura di un computer moderno:

1. CENTRAL PROCESSING UNIT (CPU)
2. RANDOM ACCESS MEMORY (RAM)
3. PERSISTENT (STORAGE) MEMORY
4. INPUT / OUTPUT DEVICES.
5. INTERCONNECTION BUSES.

The Stored-Program Concept





1948:
Claude Shannon
and
Williams, Kilburn, Toothill
(The Manchester Baby)

Shannon: *The Mathematical Theory of Communication.*



- groundwork for understanding the theoretical limits of communication between people and machines
- Identified the bit as a fundamental unit of information and, coincidentally, the basic unit of computation.
- The first program, consisting of seventeen instructions and written by Kilburn, ran on June 21st, 1948.
- This was the first program in history to run on a digital, electronic, stored-program computer.



1952: Grace Murray Hopper

- La programmazione dei primi computer era DIFFICILE.
- Si programmava direttamente inserendo le istruzioni in BINARIO dentro la macchina.
- Grace Hopper propone e progetta A-0 System, un Programma che traduce da un linguaggio formale ad alto livello nelle sequenze binarie.
- E' l'idea del primo COMPILATORE.
- Viene realizzato, per girare sull'UNIVAC.
- E' anche il primo software libero ed open-source nella storia dell'Informatica.
- Ebbe un ruolo primario nella progettazione del **linguaggio COBOL**. (1959).

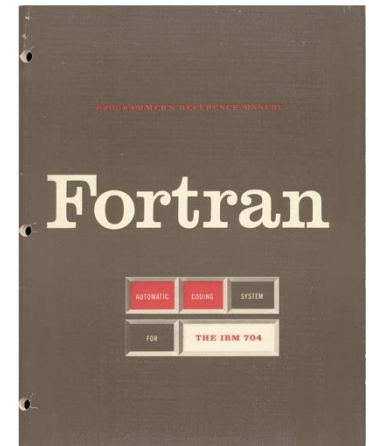
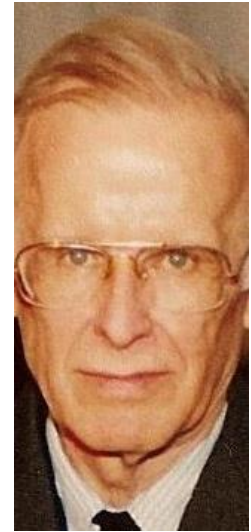
P.s. The Cathedral and The Bazaar.

Curiosità: È famosa anche per aver diffuso l'uso del termine **bug** per indicare un malfunzionamento di un computer, dopo aver estratto una falena dal Mark II

John Backus

And the Fortran Language

- **1954** - FORTRAN = Formula Translator.
- Compiled – Imperative Programming Language for Scientific Computation.
- A fundamental step in programming.
- BNF and Theory of Compiler Design.
- First Compiler in 1957.
- Billions of line of codes still in use today.
- Still used and evolving language.



https://it.wikipedia.org/wiki/John_Backus

<https://en.wikipedia.org/wiki/Fortran>

WHAT IS INTELLIGENCE?

- *"It is not my aim to surprise or shock you – but the simplest way I can summarize is to say that there are now in the world machines that can think, that learn, and that create.*
- *Moreover, their ability to do these things is going to increase rapidly until — in a visible future — the range of problems they can handle will be coextensive with the range to which human mind has been applied. "*
- **by Herbert A Simon (1957)**

Allen Newell, Herbert A. Simon and J.C. Shaw

begin work on ***Logic Theorist***, a program that would eventually prove 38 theorems from Whitehead and Russell's *Principia Mathematica*. *Logic Theorist* introduced several critical concepts to artificial intelligence including heuristics, list processing and 'reasoning as search.'

Origini dell'Intelligenza Artificiale



Conferenza di Dartmouth (1956)

- **Primo linguaggio 'simbolico', LISP = list processing.**
- Primo Linguaggio Funzionale Interpretato (o Compilato).
- Prime applicazioni di ragionamento automatico (theorem proving, game playing).
- Rappresentazione esplicita della conoscenza (logica formale).
- Sistemi Esperti.

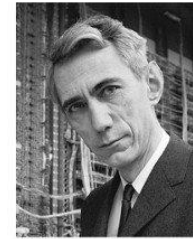
1956 Dartmouth Conference: The Founding Fathers of AI



John McCarthy



Marvin Minsky



Claude Shannon



Ray Solomonoff



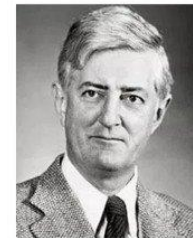
Alan Newell



Herbert Simon



Arthur Samuel



Oliver Selfridge



Nathaniel Rochester



Trenchard More

<https://it.wikipedia.org/wiki/Lisp>



NOTABLE AI MOMENTS (1940–1970)

- 1943 - McCulloch & Pitts: Boolean circuit model of brain
- 1950 - Alan Turing's "Computing Machinery and Intelligence"
- 1951 - Marvin Minsky develops a neural network machine
- 1950s Early AI programs: e.g., Samuel's checkers program, Gelernter's Geometry Engine, Newell & Simon's Logic Theorist and General Problem Solver
- 1956 - Dartmouth meeting: "Artificial Intelligence" adopted
- 1965 - Robinson's complete algorithm for logical reasoning
- 1966 - Joseph Weizenbaum creates Eliza
- 1969 - Minsky & Papert show limitations of the perceptron
Neural network research almost disappears

Modern Computing Milestones

- 1960 – Primo IC (Integrated Circuit)
- 1962 – CTSS (First Multitasking OS)
- 1965 - Moore Law
- 1969 – First version of UNIX
- 1971 - Primo microprocessore (Intel 4004)
- 1984 – IBM Personal Computer.